

Iasta: An Efficient Cipher for Hybrid Homomorphic Encryption in Cloud-Assisted IoT

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Abstract—The ciphertext expansion in fully homomorphic encryption (FHE) leads to a high client-server communication overhead. To address this, hybrid homomorphic encryption (HHE) reduces bandwidth consumption by employing a symmetric cipher for client-side encryption and converting the ciphertext to FHE format on the server side, thus enabling efficient and secure data aggregation in Internet of Things (IoT) scenarios. However, existing HHE schemes incur high computational costs on client devices and significant homomorphic conversion overhead on the server side. Designing HE-friendly symmetric ciphers with low AND depth and low randomness remains a major challenge. Therefore, this paper proposes Iasta, a novel stream cipher optimized for integer-based HHE in IoT systems. Iasta employs polynomial modular arithmetic to support a non-binary plaintext space and utilizes rotation-based linear transformations to generate random matrices, reducing the randomness required in the affine layer. Furthermore, by replacing the original non-linear layer with a dual S-box design, the cipher reduces the number of rounds while maintaining the same security level. Finally, we implemented a branch-parallel homomorphic evaluation of Iasta using layered and inverse rotation techniques. In the SEAL library, experimental comparisons with Pasta, the fastest known implementation over fields $\mathbb{F}_p$, demonstrate that Iasta achieves a $2\times$ higher client-side throughput at half the storage cost, along with $3 - 5\times$ higher throughput and $2\times$ faster runtime on the server side.

Index Terms—Hybrid homomorphic encryption, Internet of Things, homomorphic conversion, branch-parallel homomorphic evaluation.

I. INTRODUCTION

WITH the rapid growth of Internet of Things (IoT) applications, data volumes have increased exponentially, while resource-constrained edge devices face challenges in efficiently processing large-scale datasets. To support scalable deployment, IoT leverages cloud-assisted outsourcing services for data management. However, uploading data to the cloud raises privacy concerns, while encrypting data makes it difficult for the cloud

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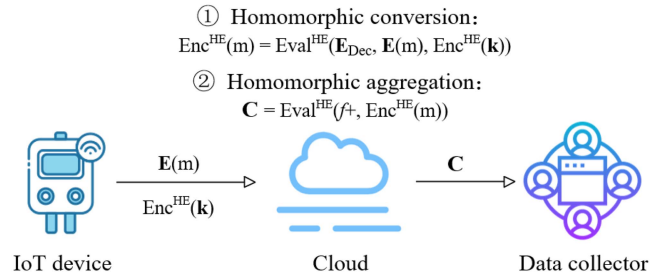


Fig. 1. HHE scheme applied to cloud-assisted IoT.

to perform direct statistical analysis. Fully homomorphic encryption (FHE) enables computation on encrypted data without decryption, thus balancing privacy protection and computational needs. Therefore, integrating cloud-assisted IoT with homomorphic encryption (HE) presents a promising solution to meet the increasingly stringent privacy requirements in IoT scenarios.

Unfortunately, a single HE scheme can hardly adapt to IoT scenarios: first, it introduces computational delays during data acquisition and transmission, and second, the ciphertext generated by HE is several orders of magnitude larger than the corresponding plaintext (i.e., ciphertext expansion). In resource-constrained IoT environments with limited bandwidth, storage, and computational capabilities, ciphertext expansion leads to significant performance penalties.

To address these challenges, hybrid homomorphic encryption (HHE) was first proposed in [1]. As shown in Fig. 1, in an HHE framework, IoT devices encrypt data using a symmetric key encryption (SKE) scheme $\mathbf{E}$ (with an expansion factor of 1), and then encrypt the symmetric key k with the public key of the data collector through an HE scheme. The cloud server can homomorphically evaluate the decryption circuit of $\mathbf{E}$, thereby converting symmetric ciphertexts into homomorphic ciphertexts (e.g., homomorphic conversion). Subsequently, the data undergo homomorphic processing f^+ and are transmitted to the data collector. The key advantage of this approach lies in offloading ciphertext expansion and computational complexity from resource-constrained IoT devices to cloud. Specifically, IoT devices only need to perform homomorphic encryption on short symmetric keys. This architecture significantly mitigates the issue of ciphertext expansion, and hybrid HE [2], [3] is widely considered a theoretically optimal solution for privacy-preserving data aggregation. However, these benefits come at the cost of computational overload in the cloud, which homomorphically evaluates symmetric decryption circuits. Given that a cloud server is inherently

more powerful than edge devices, this trade-off is worth considering in practice. Yet, it is crucial to deliberate on how to design HE-friendly symmetric ciphers with fixed and low AND-depth.

A. Related Work

Typically, researchers utilize AES as a symmetric cipher [4] in the HHE scheme due to its ease of integration into IoT device chips. However, performing AES decryption circuits homomorphically requires a multiplicative depth of at least 40, thereby increasing the complexity of the cloud. Clearly, homomorphic multiplication incurs a far higher computational cost compared to homomorphic addition. In the HHE framework, more AND operations in a symmetric cipher lead to higher homomorphic decryption overhead. Therefore, classical symmetric primitives are unsuitable for this scenario, necessitating the design of symmetric ciphers with high XOR operation counts and low AND operation requirements to enhance decryption performance under FHE.

The design of HE-friendly symmetric ciphers follows two principles: a fixed and sufficiently low AND depth and a minimized number of ANDs per encrypted bit. LowMC [5] is the first construction based on design principles. LowMC achieves a remarkably low number of AND operations per bit (3 to 4), but it suffers from an AND depth of at least 12 and is vulnerable to algebraic attacks. Similarly, Kreyvium [6] optimizes the number of ANDs per bit, but its AND depth increases proportionally to the length of the output keystream. In contrast, FLIP [7] maintains a low AND depth of 4 for keystream generation, but requires an excessive number of ANDs per bit (1072 ANDs/bit). A breakthrough occurred in 2018 with Rasta [10], which simultaneously minimized both metrics for the first time. This scheme adopts an ASASA-like structure (alternating key-dependent affine layers and key-dependent S-boxes), delivering the minimal ANDs per bit and the lowest AND depth (both identical and only 2 to 6), thereby establishing itself as the state-of-the-art HE-friendly stream cipher family. Rasta has spawned multiple enhanced variants (Dasta [11], Masta [13], Fasta [12], Pasta [14], [15], etc.): Dasta [11] reduces randomness requirements by streamlining Rasta's linear layer; Fasta [12] employs a 5-parallel architecture with a Helib parameter-optimized linear layer, achieving $25\times$ acceleration in packed evaluation; Masta [13] and Pasta [14] optimize Rasta by extending binary operations to multi-ary versions, achieving a thousandfold improvement in terminal throughput and server computational efficiency. Symmetric primitives constructed over the prime field $\mathbb{F}_p$ [16], [17] offer superior efficiency and are becoming the preferred choice for FHE schemes.

However, Rasta and its variants are designed primarily for second-generation FHE schemes (such as BGV [18] and BFV [19], [20]) for encrypting integer plaintexts, making them difficult to adapt to other schemes (e.g., CKKS [21] and TFHE [22]). In [8], HERA is combined with BFV to perform homomorphic translation, followed by converting the BFV ciphertexts into the CKKS ciphertexts. In contrast, Rubato [9], specifically designed for CKKS, significantly reduces the number of rounds in the stream cipher by introducing the same

approximation error, but is incompatible with other HE schemes. Moreover, researchers have introduced efficient symmetric ciphers (e.g., Elisabeth [23], FRAST [24], Thunderbird [25]) for updated TFHE schemes, primarily focusing on optimizing the number of gates in TFHE's gate-bootstrapping mode. However, the original TFHE scheme [22] only supports boolean plaintexts, and its optimized version [26] only enables low-precision integer computations.

As discussed above, each category of HE-friendly ciphers suffers from specific limitations in terms of AND depth, ANDs per bit, FHE compatibility, or randomness overhead. Table I summarizes these key limitations for representative ciphers. LowMC and Kreyvium suffer from excessive or unbounded AND depth. FLIP achieves shallow depth but incurs prohibitive ANDs per bit. The Rasta family, while balancing depth and AND count, is limited to fast evaluation on specific FHE libraries and incurs significant randomness overhead. Rubato and TFHE-oriented ciphers are locked into CKKS or TFHE, lacking cross-scheme portability. HERA enables hybrid use but adds conversion overhead. Overall, no existing cipher simultaneously meets the demands of low AND depth, minimal ANDs per bit, and broad FHE compatibility. This paper focuses on bridging this gap by designing a cipher that matches the best-in-class AND depth and ANDs per bit, while achieving minimal randomness overhead and fast homomorphic conversion.

B. Motivation

From the perspective of HE-friendly symmetric cipher design principles, stream ciphers like Rasta have established the dominant research direction in the HHE framework, and subsequent works are built upon similar ASASA-like structures. Although the primary advantage of this architecture lies in its fixed AND depth (dependent solely on the cipher iteration rounds), three inherent limitations hinder its practical applicability.

1) *Inefficiency of Integer Arithmetic*: Most HE-friendly symmetric ciphers are based on Boolean circuits. However, when dealing with statistics or integer arithmetic, the efficiency of their symmetric encryption diminishes considerably, requiring the development of binary circuits with significantly greater multiplication depth to perform integer arithmetic.

2) *High Randomness in the Affine Layer*: The affine layer in Rasta-type stream ciphers adopts the form $\mathbf{F} = \mathbf{M} \cdot \mathbf{x} + \mathbf{b}$, where $\mathbf{x}$ denotes the secret vector, while both the matrix $\mathbf{M}$ and the constant vector $\mathbf{b}$ are generated from a substantial pool of random numbers, leading to significant storage and computational overhead on client-side devices. A critical research issue lies in reducing the randomness embedded within the affine layer while ensuring resistance against algebraic attacks and linear attacks.

3) *Incompatibility Challenges Across Diverse HE Libraries*: Both the BGV and BFV schemes support ciphertext with multiple slots, and critical divergences emerge: BGVrns¹ restricts the ciphertext slots to even numbers, whereas BFV requires the slot count to be a power of two. Notably, odd-bit slot configurations

¹ See [27] for a discussion on the very similar BFVrns scheme

TABLE I
COMPARISON OF HE-FRIENDLY SYMMETRIC CIPHERS (SECURITY LEVEL $\lambda = 128$ BIT)

Cipher	AND Depth	ANDs per bit	Plaintext	FHE Compatibility	Randomness	Homomorphic Conv. Speed
LowMC [5]	≥ 12	3–4	$\mathbb{F}_2$	BGV, BFV	Low	Slow
Kreyvium [6]	≥ 12	2–3	$\mathbb{F}_2$	BGV, BFV	Low	Slow
FLIP [7]	4	1072	$\mathbb{F}_2$	BGV	Low	Very Slow
HERA [8]	5–6	5–6 / $\lceil \log p \rceil$	$\mathbb{F}_p$	BFV, CKKS	Medium	Moderate
Rubato [9]	5	2–4 / $\lceil \log p \rceil$	$\mathbb{F}_p$	CKKS	Low	Fast
Rasta/Dasta [10], [11]	4–6	4–6	$\mathbb{F}_2$	BGV, BFV	High	Moderate
Fasta [12]	6	6	$\mathbb{F}_2$	BGV	Minimal	Fast
Masta/Pasta [13], [14]	4–5	4–5 / $\lceil \log p \rceil$	$\mathbb{F}_p$	BGV, BFV	Low	Moderate
Ours	4–5	4–5 / $\lceil \log p \rceil$	$\mathbb{F}_p$	BGV, BFV	Minimal	Fast

[†] TFHE gate-bootstrapping ciphers (e.g., Elisabeth, FRAST, Thunderbird) are excluded, as their evaluation metrics are incomparable with AND depth.

are exclusively supported in the HELib implementation of BGV, mirroring the invertibility condition of S-boxes in Rasta under odd-bit configurations. Furthermore, the CKKS scheme introduces inherent approximation errors but enables native floating-point arithmetic. In contrast, the TFHE library achieves the fastest single-bit bootstrapping efficiency at the expense of parallel processing capabilities. These heterogeneous constraints across different FHE schemes pose significant challenges in designing HE-friendly symmetric ciphers that are adaptable to various scenarios.

C. Our Contribution

To address these issues, this paper proposes Iasta, which operates over a finite field $\mathbb{F}_p$ to enable large-integer arithmetic. By sampling elements from $\mathbb{F}_p$, Iasta reduces the random bits required for the matrix $\mathbf{M}$ in the affine layer. In addition, it adopts an optimized S-box configuration to achieve equivalent security with the minimum number of rounds. The contributions of this paper are as follows.

- *Designing HHE over $\mathbb{F}_p$* : Iasta has a plaintext space of $\mathbb{F}_p^n$, processing 2 polynomial rings per encryption. The linear transformation of its affine layer is computed modulo R_p , where $R_p = \mathbb{F}_p[X]/(X^n + 1)$ denotes the polynomial ring and n represents the number of ciphertext slots (without parity constraints). This modular arithmetic framework enables Iasta to seamlessly integrate with BGV- or BFV-style HE schemes, thereby achieving adaptability to diverse complex scenarios in IOT environments.
- *Reducing random bits and rounds*: The substantial random bits utilized in the affine layer are inherently linked to the state size n of the cipher and the number of rounds d . To counter algebraic attacks, we employ a hybrid utilization of Feistel S-boxes and Cube S-boxes to elevate the algebraic degree of the symmetric cipher. The Cube S-box provides an additional algebraic degree while simultaneously reducing both the number of rounds and the state size in Iasta. Fewer rounds and a compact state size collectively minimize the randomness embedded within the affine layer. Moreover, thanks to the synergy between the rotation-based affine layer and the hybrid S-box, Iasta supports a longer keystream than Pasta for the same number of rounds, offering a higher security level. For the same state size, Iasta enables additional rounds, leading to enhanced algebraic security.

- *Parallelizing branch evaluation*: We split the cipher state into two parallel branches and achieved a branch-parallel architecture via homomorphic rotations, thus improving homomorphic conversion efficiency. Under identical experimental conditions, we compare the performance of three $\mathbb{F}_p$ -based ciphers: Iasta, Pasta, and Masta. When tested in the SEAL library with a 17-bit prime p , Iasta-3 is $2\times$ faster than Pasta-3 and $9\times$ faster than Masta-4. In terms of noise consumption, Iasta-3 exhibits a 20% reduction versus Pasta-3 and 40% lower noise than Masta-4. Moreover, Iasta-3 shows $5\times$ higher throughput than Pasta-3 and $6\times$ over Masta-4. These performance advantages become more pronounced with increasing plaintext modulus and additional cipher rounds, as clearly illustrated in Fig. 6, where Iasta consistently outperforms Pasta and Masta across all evaluated metrics.

This paper is organized as follows. Section II describes the related cryptographic algorithms. In Section III, we propose a new stream cipher, Iasta, and in Section IV, we describe an efficient HHE scheme for IOT in detail. In Section V, we contrast the concrete parameters and show the corresponding experimental results. Finally, a conclusion is given in Section VI.

II. PRELIMINARIES

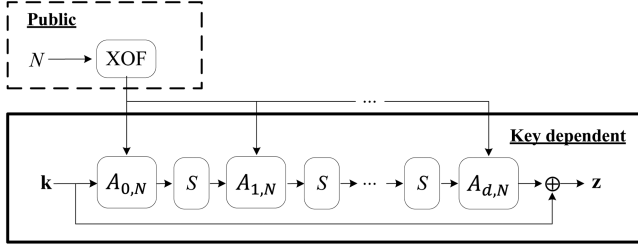
A. Notation

In this paper, vectors are denoted in lowercase bold (e.g., $\mathbf{k} \in R_p^2$), matrices in uppercase bold (e.g., $\mathbf{M} \in \mathbb{Z}_p^{n \times n}$), and algorithm names in uppercase bold.

B. BFV Scheme

We briefly introduce the BFV scheme [19], [20], which is based on the ring learning with errors (RLWE) problem. The plaintext space is taken as R_p for some integer $p \geq 1$, and the encryption scheme is defined as follows:

- **BFV.KGen**(λ, L).
 - Given the security parameter λ and the level parameter L , set $q_\ell = 2^\ell$ for $1 \leq \ell \leq L$. Let $\Delta = \lfloor q_\ell/p \rfloor$.
 - Sample $s \leftarrow R_2$, $a \leftarrow U(R_{q_L})$, and $e \leftarrow \chi$, set the secret key $\mathbf{sk} \leftarrow (1, s)$, set $\mathbf{pk} \leftarrow (a, b = -as + e) \in R_{q_L}^2$.
 - Sample $a' \leftarrow U(R_{P \cdot q_L})$ and $e' \leftarrow \chi$. Set $\mathbf{evk} \leftarrow (a', b' = -a's + e' + P \cdot s^2) \in R_{P \cdot q_L}^2$.

Fig. 2. The d -round Rasta construction.

- **BFV.Enc(pk, m)**: To encrypt a message $m \in R_p$, sample $v, e_0, e_1 \leftarrow \chi$, output $\mathbf{c} \leftarrow v \cdot \mathbf{pk} + (\Delta \cdot m + e_0, e_1) \in R_{q_\ell}^2$.
- **BFV.Dec(sk, c)**: For a ciphertext $\mathbf{c} = (c_0, c_1) \in R_{q_\ell}^2$, compute $\langle \mathbf{c}, \mathbf{sk} \rangle = c_0 + c_1 \cdot s \bmod q_\ell$, output $m = \left\lfloor \frac{p \cdot \langle \mathbf{c}, \mathbf{sk} \rangle}{q_\ell} \right\rfloor \bmod p$.
- **BFV.Add(c, c')**: For $\mathbf{c}, \mathbf{c}' \in R_{q_\ell}^2$, output $\mathbf{c}_{add} \leftarrow \mathbf{c} + \mathbf{c}' \bmod q_\ell$.
- **BFV.Mult(evk, c, c')**: For $\mathbf{c}, \mathbf{c}' \in R_{q_\ell}^2$, compute $(d_0, d_1, d_2) = \left(\left\lfloor \frac{p \cdot c_0 c'_0}{q_\ell} \right\rfloor, \left\lfloor \frac{p \cdot (c_0 c'_1 + c_1 c'_0)}{q_\ell} \right\rfloor, \left\lfloor \frac{p \cdot c_1 c'_1}{q_\ell} \right\rfloor \right)$ and output $\mathbf{c}_{mult} \leftarrow (d_0, d_1) + \lfloor P^{-1} \cdot d_2 \cdot evk \rfloor \bmod q_\ell$.

C. Rasta Stream Cipher

Rasta [10] takes the secret key $\mathbf{k} \in \mathbb{Z}_2^n$ as input and completes the permutation process by interleaving the affine layer $A_{i,N}$ and the non-linear layer S over d rounds, in which $0 \leq i \leq d$. The affine layer is generated by an extendable-output function (XOF), which takes a nonce N as input and is not reused with the same key.

As shown in Fig. 2, the keystream $\mathbf{z} \in \mathbb{Z}_2^n$ is generated as follows: $\mathbf{z} = \text{Rasta}^d[\mathbf{k}, N] = \mathbf{k} \oplus (A_{d,N} \circ S \circ A_{d-1,N} \circ \dots \circ S \circ A_{0,N})(\mathbf{k})$.

Non-linear layer S : Rasta employs the χ transformation [] as the non-linear layer, which is invertible only when n is an odd number. For a given input $\mathbf{x} = (x_0, x_1, \dots, x_{n-1}) \in \mathbb{Z}_2^n$, the output of $\chi(\mathbf{x})$ is defined as follows: $\chi(\mathbf{x})_\ell = x_\ell \oplus x_{\ell+2} \oplus x_{\ell+1}x_{\ell+2}$, where $0 \leq \ell < n$.

Affine layers $A_{i,N}$: The output of the affine layer is obtained by multiplying the input $\mathbf{x} \in \mathbb{Z}_2^n$ with an $n \times n$ matrix $\mathbf{M}_{i,N}$ and subsequently adding a round constant $\mathbf{b} \in \mathbb{Z}_2^n$, that is, $A_{i,N}[\mathbf{x}] = \mathbf{M}_{i,N} \cdot \mathbf{x} \oplus \mathbf{b}$.

Note that both the matrix $\mathbf{M}_{i,N}$ and the round constant are derived from the XOF output, dependent on the public nonce N . Crucially, the matrix $\mathbf{M}_{i,N}$ must be invertible, which accounts for more than half of Rasta's encryption time and necessitates that the XOF (typically SHAKE256 [28]) generates an increased amount of randomness.

III. A NEW STREAM CIPHER: IASTA

As variants of Rasta [10], Fasta [12] and Pasta [14] represent the fastest implementations for fields $\mathbb{F}_2$ and $\mathbb{F}_p$, respectively. Fasta utilizes rotation-based linear transformations to enable efficient homomorphic evaluation, though constrained by fixed

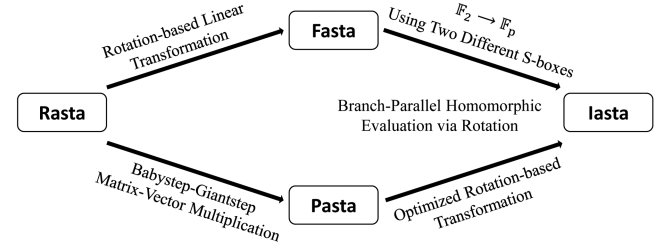


Fig. 3. The technical roadmap for the design of our scheme.

parameters only compatible with Helib. Conversely, Pasta serves as a highly optimized cipher for integer-based HHE, but retains significant randomness in its affine layer. In contrast, Iasta employs layered and inverse rotation techniques to achieve a branch-parallel architecture, unifying the circular shift operations in the rotation-based affine layer with the shift operations in the hybrid S-box. Under this architecture, we achieve efficient and parallelized homomorphic evaluation, while further reducing the randomness overhead in the affine layer.

Our design philosophy is illustrated in Fig. 3. Compared to Fasta, our cipher eliminates the dependency on HELib-specific parameters and achieves comparable or better performance across multiple HE libraries (SEAL, HELib). Compared to Pasta, Iasta significantly reduces the randomness overhead in the affine layer by unifying rotation operations, leading to lower noise growth, higher throughput, and more regular hardware implementation. The branch-parallel architecture further enables SIMD-friendly software evaluation and demonstrates good potential for hardware acceleration due to its reliance on cyclic shifts, branch-parallel additions, and multiplications.

A. Design Rationale

Applying the $\mathbb{F}_p$ stream cipher: In IoT applications where statistical and integer arithmetic operations predominate, $\mathbb{F}_p$ ciphers demonstrate superior compatibility with integer-based plaintext HE schemes (e.g., BGV, BFV). This approach not only improves client-side throughput, but also achieves comparable security levels with shorter bit-width relative to $\mathbb{Z}_2$ ciphers.

Using two different S-boxes: Both Rasta and Fasta, operating over $\mathbb{Z}_2$, utilize the χ transformation as their S-box layer. Notably, Masta, a $\mathbb{F}_p$ -variant of Rasta, has not yet addressed optimizations for the non-permutation property of the χ transformation over $\mathbb{F}_p$. Furthermore, as demonstrated in [29], researchers successfully bypassed the final round χ transformation in Rasta and Fasta by analyzing the input-output bit relationships of the χ transformation, effectively halving the security level against algebraic attacks. Therefore, we propose replacing the χ -transformation with an iterative structure comprising $d - 1$ Feistel S-boxes and a Cube S-box of higher degree, reducing the required number of rounds and appropriately minimizing the state size of the cipher. Compared to Fasta, our framework achieves accelerated homomorphic evaluation on the server side while maintaining a lower noise consumption. When compared with Pasta (another hybrid S-box design), Iasta supports a longer

keystream under the same number of rounds, while its algebraic security grows with the state size.

Reducing Randomness in the Affine layer: Generating invertible matrices in the affine layer incurs substantial computational overhead. For an $n \times n$ matrix: Rasta requires $n^2 + 3$ random bits and performs one matrix multiplication; Masta selects n random elements from $\mathbb{F}_{p^n}$ and performs $n - 1$ multiplications; Pasta samples n random elements from $\mathbb{F}_p$ and performs $n(n - 1)$ field multiplications and $(n - 1)^2$ field additions. In contrast, Fasta generates pseudorandom bits for invertible matrices based solely on the number of input words t and the iteration count r of its rotation-based linear transformation. In this work, Iasta reduces the state size of the cipher by combining a finite field $\mathbb{F}_p$ with S-boxes, thus decreasing t and r . Comprehensive cryptanalysis demonstrates that the weak randomness utilized in Iasta does not significantly compromise overall security.

Branch-parallel homomorphic evaluation via rotation: The matrix-vector multiplication in affine layers poses a major challenge to efficient homomorphic evaluation of HE-friendly ciphers. To address this, Pasta improves efficiency employing the Babystep-Giantstep algorithm for matrix-vector multiplication. However, the cipher is split into two blocks, both evaluated homomorphically, causing the runtime to scale with the cipher's state size. In contrast, Fasta decomposes the matrix-vector operation through multi-word inputs, rotations, and accumulations. It enables parallel processing by packing multiple slots into a single ciphertext. However, this method is only feasible for fast packed evaluation under specific HElib parameters, as most HE libraries do not support packing for $\mathbb{Z}_2$ ciphers. In this work, Iasta optimizes the matrix structure of the affine layer through distinct homomorphic rotation patterns, decomposing matrix-vector multiplication to enable branch-parallel homomorphic evaluation within a single cipher state.

B. Specification

In this section, we define a single instantiation of Iasta with parameters chosen to achieve 128-bit security. For a prime p , $\gcd(p - 1, 3) = 1$, a **Iasta** encryption is defined as follows:

- **KGen**(λ): Using 2 rings as the secret key, set $\mathbf{k} = (k_0, k_1) \in R_p^2$, where R_p is the ring $\mathbb{F}_p[X]/(X^n + 1)$.
- **Enc**($\mathbf{k}, \mathbf{m}, N$): Given a public nonce N , calculate a keystream $\mathbf{z} = \text{Iasta}^d[\mathbf{k}, N] \in R_p^2$. To encrypt the message $\mathbf{m} = (m_0, m_1) \in R_p^2$, output $\mathbf{c} \leftarrow \mathbf{m} + \mathbf{z}$.
- **Dec**($\mathbf{k}, \mathbf{c}, N$): For a ciphertext $\mathbf{c} = (c_0, c_1) \in R_p^2$, return $\mathbf{m} \leftarrow \mathbf{c} - \mathbf{z} = \mathbf{c} - \text{Iasta}^d[\mathbf{k}, N]$.

As shown in Fig. 4, the permutation $\text{Iasta}^d[\mathbf{k}, N]$ is defined as: $\mathbf{z} = \mathbf{k} \oplus (A_{d,N} \circ S_c \circ A_{d-1,N} \circ S_f \circ \dots \circ S_f \circ A_{0,N})[\mathbf{k}]$. Let $\mathbf{k} = (k_0, k_1) \in R_p^2$ be the input, where $k_i(X) = a_{n-1}X^{n-1} + \dots + a_2X^2 + a_1X + a_0$.

- **Non-linear layer:** The non-linear layer of Iasta is composed of $d - 1$ Feistel S-boxes and a single high-degree Cube S-box.
- S_f is the Feistel-like S-box defined as :

$$[S_f(k)]_\ell = \begin{cases} a_\ell & \text{if } \ell = 0 \\ a_\ell + a_{\ell-1}^2 & \text{otherwise} \end{cases}, \quad (1)$$

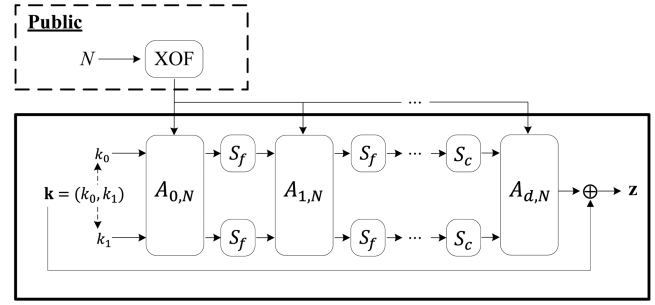


Fig. 4. The d -round Iasta construction.

where $\ell \in \{0, 1, \dots, n - 1\}$.

- S_c is the Cube S-box defined as $S_c(k)_\ell = a_\ell^3$.
- **Affine layers:** The affine layer of Iasta is implemented via a rotation-based linear transformation $\mathcal{L}$ and round constants $\mathbf{b}_{i,N} \in R_p^2$, where $A_{i,N}[\mathbf{k}] = \mathcal{L}(\mathbf{k}) + \mathbf{b}_{i,N} = \mathbf{M}_{i,N} \cdot \mathbf{k} + \mathbf{b}_{i,N}$. The matrix $\mathbf{M}$ is constructed through iterative applications of the Θ -function and word rotations, and is defined as:

$$\mathbf{M} = \mathbf{P}_{\Theta_r} \cdot \mathbf{R}_{v_{r-1}} \cdot \dots \cdot \mathbf{R}_{v_2} \cdot \mathbf{P}_{\Theta_2} \cdot \mathbf{R}_{v_1} \cdot \mathbf{P}_{\Theta_1}, \quad (2)$$

where the iteration count is r , depending on the polynomial ring R_p . We define a 2×2 matrix whose elements depend on a polynomial $p_\Theta = X^{\alpha_3} + X^{\alpha_2} + X^{\alpha_1} + 1$, where $\alpha_3 > \alpha_2 > \alpha_1$.

$$\mathbf{P}_\Theta = \begin{bmatrix} p_\Theta + 1 & p_\Theta \\ p_\Theta & p_\Theta + 1 \end{bmatrix}. \quad (3)$$

Likewise, the rotation matrix is defined as

$$\mathbf{R}_v = \begin{bmatrix} X^{\mu_0} + 1 & X^{\mu_1} \\ X^{\mu_1} & X^{\mu_0} + 1 \end{bmatrix}. \quad (4)$$

The above constructions ensure that the matrices $\mathbf{P}_\Theta$, $\mathbf{R}_v$, and the resulting matrix $\mathbf{M}$ are invertible. Moreover, word-wise rotation is implemented through matrices $\mathbf{P}_\Theta$ and $\mathbf{R}_v$ to enhance the required avalanche effect in the affine layer. The cipher guarantees that each word carrying message bits achieves full diffusion (influencing the entire state) within a few rounds. Formally, every output bit can be expressed as complex expressions of all input bits.

C. An Efficient Instance

In this section, we propose a 4-round instance of Iasta-4 with 128 bits of security for the prime fields $\mathbb{F}_p$. As shown in Table II, Iasta-4 takes as input 2 rings $\mathbf{k} = (k_0, k_1)$, sets $k_i(X) = a_{63}X^{63} + \dots + a_2X^2 + a_1X + a_0$, where $a_{j \in [64]} \in \mathbb{F}_p$.

Non-linear layer: The Feistel S-box and Cube S-box are no longer constrained by the parity of length n . We only need to set $p = 65537$, ensuring that $\gcd(p - 1, 3) = 1$ guarantees the invertibility of the Cube S-box.

Affine layers. Let $\mathbf{k} = (k_0, k_1) \in R_p^2$ be the input of a rotation-based linear transformation $\mathcal{L}$. In this case, $\mathbf{M} = \mathbf{P}_{\Theta_3} \cdot \mathbf{R}_{v_2} \cdot \mathbf{P}_{\Theta_2} \cdot \mathbf{R}_{v_1} \cdot \mathbf{P}_{\Theta_1}$. Set $k = k_0 + k_1$, the specific implementation

TABLE II
AN EFFICIENT INSTANCE IASTA-4

Symbol	Parameter	Value
λ	Security (bit)	128
d	Iasta round	4
t	Rings size	2
p	Plaintexts size	$2^{16} + 1 = 65537$
n	Degree of R_p	64
r	Iteration count of Θ	3
p_Θ	a polynomial $X^{\alpha_3} + X^{\alpha_2} + X^{\alpha_1} + 1$	$1 \leq \alpha_1 \leq 3,$ $4 \leq \alpha_2 \leq 6,$ $\alpha_3 = 7 + (2\alpha_1 + \alpha_2 + 1 \bmod 3)$
μ	word rotations in R_{v_1} and R_{v_2}	$\mu_0^{(1)} = 5 + i_1, \mu_1^{(1)} = 10 + i_2,$ $0 \leq i_1, i_2 \leq 4,$ $\mu_0^{(2)} = 14 + j_1, \mu_1^{(2)} = 28 + j_2,$ $0 \leq j_1, j_2 \leq 13.$

proceeds as follows:

$$\begin{aligned} \mathbf{P}_{\Theta_1} \cdot \mathbf{k} &= \begin{bmatrix} p_{\Theta_1} + 1 & p_{\Theta_1} \\ p_{\Theta_1} & p_{\Theta_1} + 1 \end{bmatrix} \cdot \begin{bmatrix} k_0 \\ k_1 \end{bmatrix} \\ &= \begin{bmatrix} k_0 + k \cdot p_{\Theta_1} \\ k_1 + k \cdot p_{\Theta_1} \end{bmatrix} = \begin{bmatrix} w_0 \\ w_1 \end{bmatrix}. \end{aligned} \quad (5)$$

After the first Θ -function, the number of words affected per ring (i.e. w_0, w_1) is calculated as $1 + 9 = 10$.

$$\begin{aligned} \mathbf{R}_{v_1} \cdot \mathbf{P}_{\Theta_1} \cdot \mathbf{k} &= \begin{bmatrix} w_0 + w_0 \cdot X^{\mu_0^{(1)}} + w_1 \cdot X^{\mu_1^{(1)}} \\ w_1 + w_0 \cdot X^{\mu_1^{(1)}} + w_1 \cdot X^{\mu_0^{(1)}} \end{bmatrix} \\ &= \begin{bmatrix} w_0 \\ w_1 \end{bmatrix} + \begin{bmatrix} w_0 \cdot X^{\mu_0^{(1)}} \\ w_1 \cdot X^{\mu_0^{(1)}} \end{bmatrix} + \begin{bmatrix} w_1 \cdot X^{\mu_1^{(1)}} \\ w_0 \cdot X^{\mu_1^{(1)}} \end{bmatrix} \\ &= \begin{bmatrix} w'_0 \\ w'_1 \end{bmatrix}. \end{aligned} \quad (6)$$

Following multiplication with $\mathbf{R}_{v_1}$, each ring w_t will undergo a rotation $\ll (t+1) \cdot \frac{10}{2} + i$. Therefore, the rotations before the next iteration are given as follows:

$$\begin{cases} w'_0 = w_0 + (w_0 \ll 5 + i_1) + (w_1 \ll 10 + i_2) \\ w'_1 = w_1 + (w_1 \ll 5 + i_1) + (w_0 \ll 10 + i_2) \end{cases}, \quad (7)$$

where $0 \leq i_1, i_2 \leq 4$. Then, set $w' = w'_0 + w'_1$, the number of words affected in the ring w' will sum up to at least $10 + 10 = 20$. Subsequently, after the second iteration, the minimum affected element count per input ring will reach $20 + 9 = 29$. Following multiplication with $\mathbf{R}_{v_2}$, each ring will undergo a rotation $\ll (t+1) \cdot \lfloor \frac{29}{2} \rfloor + j$. The rings are then rotated by

$$\begin{cases} w''_0 = w'_0 + (w'_0 \ll 14 + j_1) + (w'_1 \ll 28 + j_2) \\ w''_1 = w'_1 + (w'_1 \ll 14 + j_1) + (w'_0 \ll 28 + j_2) \end{cases}, \quad (8)$$

where $0 \leq j_1, j_2 \leq 13$. Upon entering the third Θ -function, the rings undergoes subsequent summation and multiplication of p_{Θ_3} . This processing ultimately results in the number of affected words at least $29 + 28 + 9 = 66$.

Given the length $n = 64$ of the ring, the entire state will be fully affected after the input undergoes 3 iterations. Any modification to an input element invariably induces distinct variations

in the output elements, thus achieving the desired avalanche effect characteristic of secure cryptographic transformations.

Limited randomness in the rotation-based affine layer: It is evident that in the affine layer of Iasta-4, the randomness used to generate the matrix $\mathbf{M}$ is fixed and only $(\alpha_1^{(r)}, \alpha_2^{(r)})$, (i_1, i_2) and (j_1, j_2) are required for three iterations. All randomness is derived from the output of an XOF(N), where the XOF is the SHAKE256 function and N is a public nonce. Consequently, each matrix of the affine layer requires 22 bits (i.e. $3^3 \cdot 3^3 \cdot 5^2 \cdot 14^2 \approx 2^{21.77}$), while the constants $\mathbf{b}_{i,N}$, consistent with the plaintext elements, occupy $t \cdot n \cdot \lfloor \log p \rfloor = 2 \cdot 64 \cdot 16 = 2048$ bits.

Remark 1: For Iasta, d is inversely proportional to r and n . To ensure 128-bit security, when $d = 3$ (i.e. Iasta-3), the parameters are set to $r = 4$ and $n = 128$. In this case, a rotation matrix $\mathbf{R}_{v_3}$ must be added with $\mu_0^{(3)} = 33 + \epsilon_1$ and $\mu_1^{(3)} = 66 + \epsilon_2$, where $0 \leq \epsilon_1, \epsilon_2 \leq 32$.

D. Cryptanalysis

In this paper, given a fixed number of rounds, our aim is to find the minimum key length n that ensures at least λ bit security. In general, $\lambda \approx \log\{p^{2n}\}$. However, unlike classical symmetric ciphers, HE-friendly symmetric ciphers primarily aim to minimize the number of rounds to ensure a sufficiently low count of AND operations. Therefore, following the approaches of Rasta [10], Masta [13], Fasta [12], and Pasta [14], the following analyses are typically employed.

1) *Algebraic Attacks:* The Iasta cipher can be represented by a set of polynomials on $\mathbb{F}_p$. For each input ring, it generates a keystream of the same size, as defined by $k_{i \in [2]} : (a_0, a_1, \dots, a_{n-1}) \rightarrow \mathbf{z} : (z_0, z_1, \dots, z_{2n-1})$. Consequently, we derive:

$$\begin{aligned} f_0(a_0, \dots, a_{n-1}) + z_0 &= 0, \\ f_1(a_0, \dots, a_{n-1}) + z_1 &= 0, \\ &\vdots \\ f_{2n-1}(a_0, \dots, a_{n-1}) + z_{2n-1} &= 0 \end{aligned} \quad (9)$$

Due to the low multiplicative depth of HE-friendly symmetric ciphers, adversaries can launch algebraic attacks by collecting a sufficient number of equations. Specific attack types include trivial linearization and the Gröbner basis attack.

Trivial Linearization: Trivial linearization involves substituting all monomials of degree greater than 1 with new variables, ultimately yielding a solvable system of linear equations. Given n_v variables and a maximum degree δ , the number of monomials is

$$S = \sum_{i=0}^{\delta} \binom{n_v + i - 1}{i} = \binom{n_v + \delta}{\delta}. \quad (10)$$

Therefore, at most S monomials are sufficient to solve the cipher system of equations, with a computational complexity of $O(S^\omega)$, where $2 \leq \omega \leq 3$. Furthermore, an adversary can reduce the maximum number of monomials by guessing g independent

variables, thus decreasing the monomial count to

$$S_g = \sum_{i=0}^{\delta} \binom{n_v - g + i - 1}{i} = \binom{n_v - g + \delta}{\delta}. \quad (11)$$

However, such a reduction incurs a p^g increase in the complexity of linearization attacks. Consequently, the total computational complexity becomes $O(p^g \cdot S_g^\omega)$.

For Iasta-4, we have $n_v = 2n = 128$ and $\delta = 2^3 \times 3 = 24$, then $S = \binom{128 + 24}{24} \approx 2^{92}$. The complexity of this attack ($S^\omega \geq 2^{184}$) significantly exceeds the security level claimed by Iasta-4 (128-bit security). Additionally, when 8 variables are guessed, the complexity increases by a factor of $p^g = 65537^8 \approx 2^{128}$. At this point, with the number of monomials being $S_g = \binom{128 - 8 + 24}{24} \approx 2^{90}$, the complexity (i.e. $S_g^\omega \geq 2^{180}$) of solving linear systems independently still exceeds the 128-bit security level.

Gröbner basis attack: Gröbner basis attacks eliminate variables by computing the Gröbner basis of the linear system. In contrast with trivial linearization, these attacks utilize minimal data, yet exhibit slower computation speeds than data-scarce trivial linearization. Therefore, Gröbner basis attacks are only effective when the number of monomials grows excessively. According to Fasta [12] and [30], as long as $O(S^\omega)$ is sufficiently large, the system can equally resist the Gröbner basis attacks.

2) **Linear Attacks:** Linear cryptanalysis relies on a sufficient number of plaintext-ciphertext pairs and linear approximations of fixed permutations to mount attacks. However, this type of attack is ineffective against Rasta-style ciphers because each encryption employs an independent random affine layer. Essentially, recovering the affine layer key resembles solving LPN (learning parity with noise) or LWE problems. Furthermore, the presence of S-boxes inherently disrupts the linear characteristics of plaintext-ciphertext pairs. Rasta derives an upper bound on the correlation of linear approximations based on the properties of the χ transformation. Specifically, Rasta employs 6 S-box layers, while Iasta-4 consists of $2 \times 4 = 8$ applications of the S-box. After 2 rounds, the linear correlation of Iasta-4 becomes comparable to that of Rasta-6. Consequently, linear cryptanalysis proves impractical against Iasta ciphers.

It is worth noting that using only the χ transformation can be stripped by adversaries in the final round [29], thus further reducing the number of monomials; whereas the approach of employing two distinct S-boxes has been rigorously validated in Pasta. Consequently, Iasta inherits the identical security characteristics as Pasta.

3) **Other Classical Attacks:** Differential cryptanalysis, cube attacks, and integral attacks require both a fixed permutation and a large number of plaintext-ciphertext pairs. In contrast, Iasta performs encryption through non-reusable pseudorandom matrices and constants that vary per encryption. Consequently, these attacks are rendered inapplicable to Iasta.

4) **Potential Quantum Attacks:** The impact of quantum attacks on homomorphic friendly symmetric ciphers first depends on how the key is exposed in the attack scenario. In the standard

model of hybrid homomorphic encryption, the symmetric key is held by the client and transmitted to the server after being encrypted once. The server never decrypts the key in plaintext during computation. Therefore, unless a quantum attack can directly recover the plaintext symmetric key from the HHE ciphertext, Grover's algorithm and similar key search attacks lack a direct entry point in the standard HHE protocol model. However, under more aggressive attack models, for instance if an adversary can obtain ciphertexts and induce the server to decrypt them, or if the implementation suffers from side channel leakage, Grover's algorithm still poses a potential theoretical threat. The algorithm reduces the complexity of brute force key search from $O(2^n)$ to $O(2^{n/2})$. This means that to maintain 128 bits of security, the key length of symmetric ciphers should be doubled to 256 bits.

For low depth ciphers optimized for homomorphic encryption, such as Iasta, Pasta, and Rasta, their circuit structures are more regular and compact. This makes the quantum Oracle implementation in Grover's algorithm relatively efficient. This implies that for the same key length, the practical cost of a quantum attack on such ciphers may be lower than on more complex traditional ciphers such as AES. However, this does not constitute a fundamental security vulnerability. It can be effectively mitigated by appropriately increasing the key length or state size during parameter selection. More general post quantum threats remain primarily confined to Grover's algorithm, with no evidence that Shor's algorithm or its variants can be used to attack symmetric ciphers. In practical deployments, it is advisable to use higher round versions such as Iasta-4, combined with 256 bit security parameters, to balance classical homomorphic efficiency with long term security.

IV. AN EFFICIENT HHE IN CLOUD-ASSISTED IoT

A. A Hybrid Scheme of Iasta and FHE

As shown in Fig. 5, we propose an efficient HHE framework as described in the following.

- **HHE.KGen(λ):** Input the security parameter λ , do the following:
 - Run **BFV.KGen(λ, L)**, return (pk, sk, evk) .
 - Run **Iasta.KGen(λ)**, return k .
 Then, output (sk, k) and (pk, evk) .
- **HHE.Enc(m, k, pk):** Take the message m as input, do the following:
 - Run $c \leftarrow \text{Iasta.Enc}(k, m)$,
 - Run $c_k \leftarrow \text{BFV.Enc}(pk, k)$.
 output (c, c_k) .
- **HHE.Conv(evk, c, c_k):** Given a symmetric cipher c , the homomorphic conversion algorithm outputs an HE ciphertext $c' \leftarrow \text{BFV.Eval}(evk, \text{Iasta.Dec}(c, c_k))$.
- **HHE.Agg($evk, f, c'_1, \dots, c'_n$):** For n HE ciphertexts and a function f , run **BFV.Eval($evk, f, c'_1, \dots, c'_n$)**, then output the aggregated result c^* .
- **HHE.Dec(sk, c^*):** Run **BFV.Dec(sk, c^*)**, output an aggregated message m^* .

Remark 2: In our hybrid scheme, the BFV level parameter L must exceed the number of Iasta rounds d .

The security of the HHE system mainly consists of two requirements. First, the IND-CPA security of the scheme

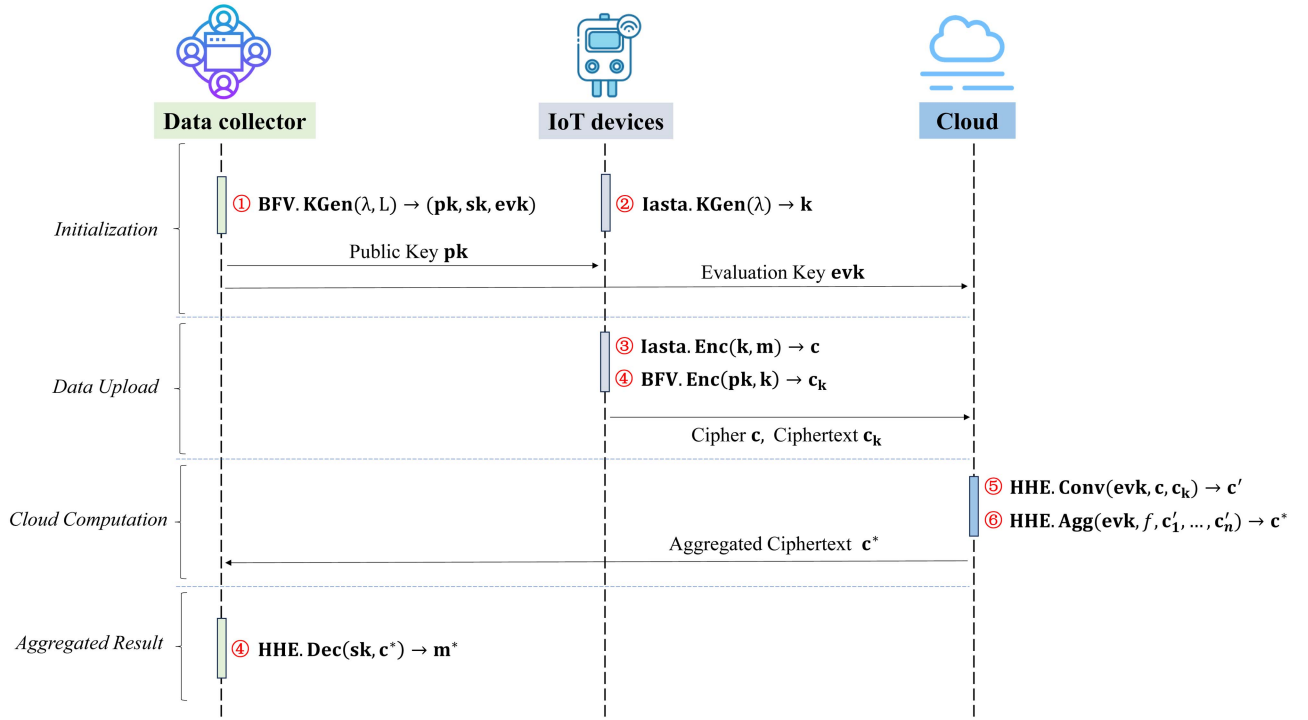


Fig. 5. A Hybrid Scheme of Iasta and BFV for cloud-assisted IoT.

itself (i.e., indistinguishability under chosen-plaintext attack), ensuring that no probabilistic polynomial-time adversary can distinguish between encryptions of different plaintexts. Second, the confidentiality of data, meaning that beyond the authorized data collector, no one should be able to access the private data of IoT devices. Therefore, our system must be able to resist the following attacks launched by malicious parties: eavesdropping attacks, where an adversary intercepts channel communications to obtain private information; and collusion attacks, where a malicious party colludes with terminal devices or the cloud to obtain other parties' private information.

B. Security Analysis

The IND-CPA security of the HHE scheme [31] is based on that of its underlying components: the stream cipher Iasta and the BFV scheme. The security of Iasta is achieved via resistance to a wide range of known cryptanalytic attacks (detailed in Section III-D), while the security of BFV is established in [20]. We therefore present the security analysis of the HHE scheme as follows.

Theorem 1: If both the Iasta and BFV schemes are IND-CPA secure, then so is their hybrid homomorphic encryption scheme.

Proof: Given a probabilistic polynomial-time adversary $\mathcal{B} = (\mathcal{B}_1, \mathcal{B}_2)$, its advantage in breaking the semantic security of the hybrid homomorphic encryption scheme is

$$\text{Adv}_{\mathcal{B}}(\lambda) = \left| \Pr[\mathcal{B}_2(x_0, x_1, y, \text{pk}, \text{evk}, \text{c}_k) = b] \right.$$

$$(\text{pk}, \text{sk}, \text{evk}) \leftarrow \text{BFV.KGen}, k \leftarrow \text{Iasta.KGen},$$

$$\begin{aligned} \text{c}_k &\leftarrow \text{BFV.Enc}, (x_0, x_1, \text{pk}_{\text{HHE}}) \leftarrow \mathcal{B}_1(\text{pk}, \text{evk}, \text{c}_k), \\ b &\leftarrow \{0, 1\}, y \leftarrow \text{Iasta.Enc}(x_b) - \frac{1}{2}. \end{aligned} \quad (12)$$

We set up two chosen-plaintext attack games for Iasta and BFV to compute the advantage of the adversary $\mathcal{B}$ to break SKE or FHE. In the SKE game, the adversary $\mathcal{A}_{\text{SKE}}$ only has access to the public nonce N , interacts with adversary $\mathcal{B}$ through the following game.

- 1) $\mathcal{A}_{\text{SKE}}$ runs $k' \leftarrow \text{Iasta.KGen}$, $(\text{pk}, \text{sk}, \text{evk}) \leftarrow \text{BFV.KGen}$, and $\text{c}_{k'} \leftarrow \text{BFV.Enc}$.
- 2) $\mathcal{A}_{\text{SKE}}$ sends $(\text{pk}, \text{evk}, \text{c}_{k'})$ as the public key pk_{HHE} of the HHE scheme to $\mathcal{B}_1$.
- 3) $\mathcal{B}_1$ sends two messages (x_0, x_1) to $\mathcal{A}_{\text{SKE}}$.
- 4) $\mathcal{A}_{\text{SKE}}$ chooses $b \leftarrow \{0, 1\}$, runs $k \leftarrow \text{Iasta.KGen}$, $y \leftarrow \text{Iasta.Enc}(k, x_b)$ and sends y to $\mathcal{B}_2$.
- 5) Adversaries $\mathcal{B}_2$ outputs $b' \in \{0, 1\}$ to $\mathcal{A}_{\text{SKE}}$.

For Adversary $\mathcal{B}$, steps 2)–5) are the same as in the standard game of the HHE scheme, except for the unmatched $\text{c}_{k'}$ and y . If the BFV scheme is semantically secure, then the distinguishing advantage between the above game and the standard game is negligible. Therefore, we can conclude that $\text{Adv}_{\mathcal{A}_{\text{SKE}}} \leq \text{Adv}_{\mathcal{B}} - \text{negl}(\lambda)$.

In the FHE game, the adversary $\mathcal{A}_{\text{FHE}}$ only has access to (pk, evk) . The game is as follows.

- 1) $\mathcal{A}_{\text{FHE}}$ runs $k \leftarrow \text{Iasta.KGen}$, $(\text{pk}', \text{sk}', \text{evk}') \leftarrow \text{BFV.KGen}$.
- 2) $\mathcal{A}_{\text{FHE}}$ runs Iasta.KGen again to get k' .

3) $\mathcal{A}_{\text{FHE}}$ sets $x_0 = \mathbf{k}$, $x_1 = \mathbf{k}'$, chooses a random bit $b \in \{0, 1\}$, then computes $y \leftarrow \text{BFV.Enc}(\mathbf{pk}', x_b)$.

4) $\mathcal{A}_{\text{FHE}}$ sends $(\mathbf{pk}', \text{evk}', y)$ to $\mathcal{B}_1$.

5) $\mathcal{B}_1$ sends x'_0, x'_1 to $\mathcal{A}_{\text{FHE}}$, then $\mathcal{A}_{\text{FHE}}$ chooses a bit $\mu \leftarrow \{0, 1\}$, runs $y' \leftarrow \text{Iasta.Enc}(\mathbf{k}, x'_\mu)$ and sends y' to $\mathcal{B}_2$.

6) $\mathcal{B}_2$ outputs $\mu' \in \{0, 1\}$ to $\mathcal{A}_{\text{FHE}}$.

7) $\mathcal{A}_{\text{FHE}}$ guesses whether $\mathcal{B}_2$ can output correctly. If $\mu' = \mu$, $\mathcal{A}_{\text{FHE}}$ outputs $b = 0$, otherwise $b = 1$.

If $b = 0$, then y is a homomorphic ciphertext of $\mathbf{k}$. In this case, steps 4)–6) are identical to the standard game, and the adversary $\mathcal{B}$ has a distinguishing advantage $\text{Adv}_{\mathcal{B}}$. If $b = 1$, the game aligns with the one we designed for the SKE scheme. Thus, the distinguishing advantage of Adversary $\mathcal{A}_{\text{FHE}}$ is

$$\begin{aligned} \text{Adv}_{\mathcal{A}_{\text{FHE}}} &= \Pr[\text{Correct guess}] - \frac{1}{2} \\ &= \Pr[b = 0 | \text{Guess } 0] + \Pr[b = 1 | \text{Guess } 1] - \frac{1}{2} \\ &= \frac{1}{2} \cdot \left(\frac{1}{2} + \text{Adv}_{\mathcal{B}}\right) + \frac{1}{2} \cdot \left(\frac{1}{2} - \text{Adv}_{\text{ASKE}}\right) - \frac{1}{2} \\ &= \frac{1}{2} \cdot (\text{Adv}_{\mathcal{B}} - \text{Adv}_{\text{ASKE}}). \end{aligned} \quad (13)$$

To sum, $\text{Adv}_{\mathcal{B}} = 2\text{Adv}_{\mathcal{A}_{\text{FHE}}} + \text{Adv}_{\text{ASKE}}$. Therefore, if either $\text{Adv}_{\mathcal{A}_{\text{FHE}}}$ or Adv_{ASKE} is negligible, so is $\text{Adv}_{\mathcal{B}}$. This completes the proof.

Next, we analyze the data confidentiality of the system. The HHE scheme itself satisfies IND-CPA security, which effectively resists eavesdropping attacks and collusion attacks.

Eavesdropping attack: Adversaries and malicious cloud servers cannot obtain the symmetric key or the homomorphic decryption key, and thus cannot extract any private information from intercepted channel messages, unless they manage to break Iasta or the BFV scheme. As a one-time-pad symmetric stream cipher, Iasta effectively resists algebraic attacks, linear attacks, and classical differential attacks. Even against stronger post-quantum attacks (e.g., Grover's algorithm), it lacks an effective point of entry without access to a decryption oracle. The BFV scheme is based on the ring learning with errors (RLWE) problem and, under practical security parameters, is widely recognized to resist known classical and quantum attacks, thereby ensuring the data confidentiality of the system.

Collusion attack: Collusion attacks similarly do not compromise the security of HHE. First, except for the authorized data collector, no malicious party can obtain the homomorphic decryption key by combining available information. Second, without the homomorphic decryption key, no party can decrypt the ciphertext that encrypts the symmetric key, i.e., they cannot obtain the symmetric key itself. Therefore, the security of HHE remains intact under collusion attacks.

C. Homomorphic Parallelizing Evaluation of Iasta

In this section, we present a branch-parallel homomorphic evaluation of the affine layer in Iasta, demonstrating efficient instantiation with parallel processing capabilities.

In RLWE-based implementations (e.g., HELib and SEAL), plaintexts are input as integer vectors and encoded into

Algorithm 1: HE.Aff(state; r; N).

Input: Homomorphic ciphertext $state \in R_p$, iteration count r , public nonce N .

Output: an value of $A_N[state]$

```

1:  $a_0 \leftarrow state$ 
2:  $b_0 \leftarrow \text{rotate\_rev}(a_0, \varphi(m) - 1)$ 
3:  $\alpha_1^{(r)}, \alpha_2^{(r)}, \mu_0^{(r-1)}, \mu_1^{(r-1)}, b_N \leftarrow \text{XOF}(N);$ 
    $\alpha_3^{(r)} = 7 + (2\alpha_1^{(r)} + \alpha_2^{(r)} + 1 \bmod 3); b_N \in R_p$ 
4: for  $n \leftarrow 0$  to  $r - 2$  do
5:    $b_{n+1} \leftarrow a_n + b_n$ 
6:    $\beta_{n+1} \leftarrow a_n + b_{n+1} + \sum_{i=1}^3 \text{rotate}(b_{n+1}, \alpha_i^{(n)})$ 
7:    $\gamma_{n+1} \leftarrow \text{rotate\_rev}(\beta_{n+1}, \varphi(m) - 1)$ 
8:    $a_{n+1} \leftarrow$ 
      $\text{rotate}(\beta_{n+1}, \mu_0^{(n)}) + \text{rotate}(\gamma_{n+1}, \mu_1^{(n)}) + \beta_{n+1}$ 
9: end for
10:  $b_{r-1} \leftarrow \text{rotate\_rev}(a_{r-1}, \varphi(m) - 1)$ 
11:  $b_r \leftarrow a_{r-1} + b_{r-1}$ 
12:  $a_r \leftarrow a_{r-1} + b_r + \sum_{i=1}^3 \text{rotate}(b_r, \alpha_i^{(r)})$ 
13:  $state \leftarrow a_r + b_N$ 
14: return  $state$ 
```

polynomials via the inverse fast Fourier transform (IFFT), where homomorphic operations on the plaintext polynomial $R_p = \mathbb{F}_p[X]/\Phi_m(X)$ correspondingly affect these vector elements. <?TeX

$$\begin{bmatrix} \vec{k}_0 \\ \vec{k}_1 \end{bmatrix} \xrightarrow{\text{encode}} K \in R_p. \quad (14)$$

<?TeX

Moreover, Galois automorphisms can be employed to rotate the encoded vectors. In SEAL/HELlib, an encoded vector can be viewed as a hypercube [32], where automorphisms rotate data along a single dimension. For the cyclotomic polynomial ring R_p , where $\Phi_m(X) = X^n + 1$, $n = \varphi(m)$ and n is a power of two, the hypercube derived from R_p corresponds to a matrix with two rows, each of size $\frac{\text{slots}}{2}$. In this case, a Galois automorphism $\tau_i : a(X) \rightarrow a(X^i)$ can perform either:

$$\tau_{3^i}(K) \xrightarrow{\text{decode}} \begin{bmatrix} \text{rot}_i(\vec{k}_0) \\ \text{rot}_i(\vec{k}_1) \end{bmatrix}, \quad \tau_{n-1}(K) \xrightarrow{\text{decode}} \begin{bmatrix} \vec{k}_1 \\ \vec{k}_0 \end{bmatrix}. \quad (15)$$

Efficient Affine layers: We now decompose the affine transformation as follows.

$$\begin{aligned} \begin{bmatrix} \vec{w}_0 \\ \vec{w}_1 \end{bmatrix} &= \mathbf{P}_{\Theta} \cdot \mathbf{k} = \begin{bmatrix} p_{\Theta} + 1 & p_{\Theta} \\ p_{\Theta} & p_{\Theta} + 1 \end{bmatrix} \cdot \begin{bmatrix} \vec{k}_0 \\ \vec{k}_1 \end{bmatrix} \\ &= \begin{bmatrix} \vec{k}_0 \\ \vec{k}_1 \end{bmatrix} + p_{\Theta} \cdot \left(\begin{bmatrix} \vec{k}_0 \\ \vec{k}_1 \end{bmatrix} + \begin{bmatrix} \vec{k}_1 \\ \vec{k}_0 \end{bmatrix} \right). \end{aligned} \quad (16)$$

Hence, we have

$$K + p_{\Theta} \cdot (K + \tau_{n-1}(K)) \xrightarrow{\text{decode}} \begin{bmatrix} \vec{w}_0 \\ \vec{w}_1 \end{bmatrix}. \quad (17)$$

Algorithm 2: HE.Iasta(k; d; c).

Input: Symmetric key $\mathbf{k} = \vec{k}_0 \parallel \vec{k}_1$, Iasta round d , Iasta cipher $\mathbf{c}$.

Output: Homomorphic ciphertext $\mathbf{c}'$ with the same plaintext as cipher $\mathbf{c}$.

```

1:  $K \in R_p \leftarrow \text{Encode}(\mathbf{k})$ 
2:  $state \leftarrow \text{HE.Enc}(K)$ 
3: for  $n \leftarrow 0$  to  $d - 1$  do
4:    $state \leftarrow \text{HE.Aff}_n(state; r; N)$ 
5:   if  $(n = d - 1)$ 
      $S_c(state)$ ;
   else
      $S_f(state)$ ;
6: end for
7:  $state \leftarrow K + \text{HE.Aff}_d(state; r; N)$ 
8:  $\mathbf{c}' \leftarrow state + \text{Encode}(\mathbf{c})$  // add cipher
9: return  $\mathbf{c}'$ 

```

Let $K' = K + \tau_{n-1}(K)$, multiplying by p_Θ represents a rotation operation applied to K' :

$$K + K' + \sum_{i=1}^3 \tau_{3\alpha_i}(K') \xrightarrow{\text{decode}} \begin{bmatrix} \vec{w}_0 \\ \vec{w}_1 \end{bmatrix}. \quad (18)$$

Then

$$\begin{aligned} \begin{bmatrix} \vec{w}_0 \\ \vec{w}_1 \end{bmatrix} &= \mathbf{R}_v \cdot \mathbf{P}_\Theta \cdot \mathbf{k} = \begin{bmatrix} X^{\mu_0} + 1 & X^{\mu_1} \\ X^{\mu_1} & X^{\mu_0} + 1 \end{bmatrix} \cdot \begin{bmatrix} \vec{w}_0 \\ \vec{w}_1 \end{bmatrix} \\ &= \begin{bmatrix} \vec{w}_0 \\ \vec{w}_1 \end{bmatrix} + \begin{bmatrix} \vec{w}_0 \ll \mu_0 \\ \vec{w}_1 \ll \mu_0 \end{bmatrix} + \begin{bmatrix} \vec{w}_1 \ll \mu_1 \\ \vec{w}_0 \ll \mu_1 \end{bmatrix}. \end{aligned} \quad (19)$$

Set $W \xrightarrow{\text{decode}} \begin{bmatrix} \vec{w}_0 \\ \vec{w}_1 \end{bmatrix}$, we get

$$W + \tau_{3\mu_0}(W) + \tau_{3\mu_1}(\tau_{n-1}(W)) \xrightarrow{\text{decode}} \begin{bmatrix} \vec{w}'_0 \\ \vec{w}'_1 \end{bmatrix}. \quad (20)$$

By iterating the above operations, the homomorphic evaluation of the affine layer is completed, as specified in the Algorithm 1.

Similarly, by iterating through the affine layers and S-boxes, the entire homomorphic evaluation of Iasta can be completed (the S-box is simple and will not be explained in detail), as shown in the Algorithm 2.

V. PERFORMANCE EVALUATION

In this section, we evaluate the performance and feasibility of the proposed scheme.

A. Comparing Iasta With Other Ciphers

We compare with the state-of-the-art scheme of the same type. Fasta, which consists of 5-parallel calls to Rasta, with a word size of 329 and 6 rounds in total. In Table III, we summarize the parameters of Fasta and Iasta in different fields.

TABLE III
PARAMETER COMPARISON BETWEEN FASTA OVER $\mathbb{F}_2$ AND IASTA OVER $\mathbb{F}_p$
(UNIT: BIT)

| Instance | Rounds | Counts | Key | Matrix | Randomness |
|----------|--------|----------|------|-----------|--------------|
| Fasta | 6 | 4 | 1645 | 63 | 11956 |
| Iasta-3 | 3 | 4 | 4096 | 29 | 16500 |
| Iasta-4 | 4 | 3 | 2048 | 22 | 10350 |

TABLE IV
PARAMETERS OF IASTA, PASTA, AND MASTA OVER $\mathbb{F}_p$ (UNIT: BIT)

| Instance | Rounds | Key | Cipher | Matrix | Randomness |
|----------|--------|------|-------------|--------|--------------|
| Iasta-3 | 3 | 4096 | 4096 | 29 | 16500 |
| Iasta-4 | 4 | 2048 | 2048 | 22 | 10350 |
| Pasta-3 | 3 | 4096 | 2048 | 4096 | 32768 |
| Pasta-4 | 4 | 1024 | 512 | 1024 | 10240 |
| Masta-4 | 4 | 2048 | 2048 | 2048 | 20480 |
| Masta-5 | 5 | 1024 | 1024 | 1024 | 12288 |

TABLE V
COMPARISON OF RUNTIME AND THROUGHPUT BETWEEN IASTA AND FASTA IN HELIB

| Cipher | d | m | p | λ | Enc
s | Conv
s | Throughput
$bits/s$ |
|----------------|-----|-------|-------|-----------|--------------|---------------|------------------------|
| Counts $r = 3$ | | | | | | | |
| Fasta-6 | 6 | 30269 | 2 | 310 | 0.154 | 28.942 | 57 |
| Iasta-4 | 4 | 65536 | 65537 | 184 | 0.034 | 7.561 | 270 |
| Iasta-5 | 5 | 65536 | 65537 | 290 | 0.039 | 12.377 | 165 |

TABLE VI
COMPARISON OF RUNTIME AND THROUGHPUT BETWEEN IASTA, PASTA AND MASTA IN SEAL ($p = 17$ BIT)

| Cipher | n | Enc | | Conv | | Throughput |
|------------------------|-------|----------------|----------------|----------------|----------------|-------------|
| | | runtime
s | noise
bit | runtime
s | noise
bit | $bits/s$ |
| $p = 65537$ (17 bit) : | | | | | | |
| Iasta-3 | 16384 | 0.009 | 365 | 1.658 | 151 | 2457 |
| Iasta-4 | 16384 | 0.007 | 365 | 1.601 | 108 | 1274 |
| Pasta-3 | 16384 | 0.010 | 365 | 4.657 | 95 | 439 |
| Pasta-4 | 16384 | 0.014 | 365 | 2.105 | 25 | 242 |
| Masta-4 | 16384 | 0.020 | 364 | 5.912 | 32 | 345 |
| Masta-5 | 32768 | 0.046 | 800 | 21.203 | 386 | 48 |

Fasta takes five 329-bit words as input. Its linear transformation requires 4 iterations, with each affine matrix consuming 63 random bits over 6 rounds, totaling $7 \times (63 + 1645) = 11956$ bits. In contrast, Iasta demonstrates a significantly reduced randomness-per-bit ratio. Taking Iasta-4 as an example: with a $128 \times 16 = 2048$ -bit input, 3 iterations, it requires only $5 \times (22 + 2048) = 10350$ random bits in total, achieving both improved terminal throughput and reduced randomness consumption, demonstrating clear advantages.

On the other hand, Pasta, Masta and Iasta, both HE-friendly ciphers over $\mathbb{F}_p$, share the same plaintext space $p = 65537$. Table IV presents their detailed parameter. The comparison instances in the table have the same AND depth, such as Iasta-3, Pasta-3, and Masta-4.

As described in Section III-A, generate an $n \times n$ matrix where both Pasta and Masta require n elements in $\mathbb{F}_p$. For

Masta-4, the input is 128 words, and the matrix for each affine layer requires $128 \times 16 = 2048$ bits. For Pasta-3, the input is 256 words, divided into upper and lower blocks. Each block undergoes a separate affine transformation, so the matrix for each affine layer requires a total of $2 \times 128 \times 16 = 4096$ bits. Furthermore, Pasta only takes the output of the previous block as the keystream, resulting in a cipher half the size of the input: $128 \times 16 = 2048$ bits. The random bits required for the matrices dominate, and as the number of rounds increases, the overall demand for randomness grows. In summary, when $d = 3$, Iasta has half the terminal storage overhead of Pasta while achieving twice the throughput.

B. Implementation

In this section, we experimentally evaluated Iasta in SEAL and HELib,² with a focus on analyzing its computational time and noise budget consumption. The main experiments for SEAL were carried out on a local virtual machine with 2 CPU cores and 12 GB RAM, hosted on a PC with an Intel Core i9-14900HX CPU and 16 GB memory. For HELib, additional compatibility tests were performed on a remote server equipped with an AMD EPYC 7763 64-core processor and 503 GiB of RAM, as the virtual machine environment did not provide sufficient memory for HELib execution.

SEAL/HELlib parameters. We set the degree of the polynomial ring $\log n \geq 15$ to ensure the security parameter $\lambda = 128$, where $\Phi_m(X) = X^n + 1$. The initial plaintext modulus is $p = 65537$ and the initial ciphertext modulus $\log q = 365$.

1) Comparing Iasta to Fasta in HELlib: First, taking Fasta as an example, which uses the same rotation-based linear transformation, Table V lists the runtime and throughput metrics of Iasta and Fasta. The throughput is calculated as the size of the cipher divided by the total time.

Iasta-5 uses the same input and affine layer iteration counts as Iasta-4, with only one additional round of affine and non-linear

layers, achieving a security level of $\lambda = \log \left(\frac{128 + 48}{48} \right)^2 \approx$

290, and has the same AND depth as Fasta-6. In contrast, Fasta is a HE-friendly cipher specifically designed for HELlib's packed evaluation, with carefully selected parameters. When $m = 30269$ and $p = 2$ are chosen, it satisfies $\phi(m) = 30268$ and $\text{ord}_m(p) = 92$ (i.e., $2^{92} \equiv 1 \pmod{m}$), resulting in a polynomial degree of $n = \phi(m)/\text{ord}_m(p) = 329$. However, compared to the efficient evaluation presented in Section IV-C, Fasta demonstrates significantly longer encryption and homomorphic conversion times within the HHE framework. Specifically, compared to Fasta's implementation in HELlib, HELlib-based Iasta-5 achieves over $2\times$ the homomorphic conversion speed of Fasta, while Iasta delivers nearly $3\times$ the throughput of Fasta. Furthermore, HELlib allocates a 569-bit noise budget to Fasta, with the homomorphic transformation consuming 260 bits. In contrast, HELlib provides Iasta-5 with a 567-bit noise budget, consuming 384 bits. Although Iasta consumes more noise in larger plaintext modulus settings, the overall noise consumption remains well

within the allocated budget, and the trade-off is justified by its substantial performance gains.

2) Iasta Versus Pasta and Masta in SEAL: In Table VI, we compare Iasta, Pasta, and Masta for 17-bit primes in the SEAL library. Furthermore, Table VII benchmarks two distinct large primes p . We analyze the runtime and remaining noise at each step of the HHE process, with the comparative results presented in Figs. 7 and 8.

As shown in Fig. 7(a), all ciphers have an AND depth of 4, thus requiring 4 homomorphic multiplications during the homomorphic conversion process, which accounts for the majority of the runtime. When a 17-bit prime number p is given, Iasta-3 is twice as fast as Pasta-3 and $9\times$ faster than Masta-4. As shown in Fig. 8(a), when the number of multiplications increases with cipher rounds, for a 17-bit prime p , Iasta-4 is 0.5 seconds faster than Pasta-4 and $11\times$ faster than Masta-5. Overall, as the number of rounds increases and the plain modulus grows larger, the time for homomorphic conversion also increases. Compared to Pasta and Masta, Iasta has an absolute advantage in terms of speed, and when p exceeds 33 bits, the runtime tends to stabilize.

Furthermore, the noise consumption during homomorphic conversion is also a crucial metric. As shown in Figs. 7(b) and 8(b), noise consumption increases as the plain modulus grows. For an AND depth of 4, Iasta-3 incurs 56 to 232 bits less noise overhead than pasta-3 and 118 to 434 bits less than Masta-4. Naturally, a larger plain space leads to a larger n , and SEAL allocates a higher noise budget, which also increases the time overhead. In comparison, as shown in Fig. 6, Iasta enables more efficient homomorphic conversion with smaller HE parameters while maintaining a lower noise consumption.

Finally, we compared the throughput of different ciphers. Under the $\mathbb{F}_p$ benchmark, ciphers with the same AND depth maintain identical security levels. As shown in Figs. 7(c) and 8(c), Iasta demonstrates significantly higher throughput than Pasta and Masta at equivalent security levels. Specifically, for 4 AND-depth, Iasta achieves $3 - 5\times$ higher throughput than Pasta and $4 - 29\times$ higher throughput than Masta. When evaluating 5 AND-depth, Iasta exhibits $3 - 25\times$ better throughput performance than Pasta and Masta.

3) Iasta Versus Pasta and Masta in HELlib: As shown in Table VIII, all ciphers are evaluated under three plaintext modulus bit-lengths (17, 33, and 60 bits) in HELlib. In terms of conversion time, Iasta-3 consistently finishes faster than Pasta-3 and Masta-4 across all modulus sizes. As the number of rounds increases, the variation in conversion time reflects the design differences among the ciphers. Iasta-4 achieves a faster conversion time at 60-bit p (34.867 s) than Iasta-3 (40.530 s) due to fewer affine layer iterations and a halved keystream length. Pasta-4 records the fastest conversion time among all ciphers (23.500 s), benefiting from a keystream that is half the length of that in Iasta-4 and fewer rotation operations. Rotation overhead is significant in HELlib, and Iasta involves more rotation operations, which increases its conversion time. Nevertheless, Iasta features a longer keystream and a larger output cipher size, maintaining the highest throughput across all modulus sizes.

Iasta-3 delivers $2 - 8\times$ higher throughput than Pasta-3 and $3 - 8\times$ higher than Masta-4 across all modulus sizes. The

²<https://github.com/IAIK/hybrid-HE-framework.git>

TABLE VII
COMPARISON OF RUNTIME AND THROUGHPUT BETWEEN IASTA, PASTA AND MASTA IN SEAL ($p = 33/60$ BIT)

| Cipher | n | Enc | | Conv | | Throughput | n | Enc | | Conv | | Throughput |
|-----------------------------|-------|----------------|----------------|----------------|----------------|------------|--------------------------------------|----------------|----------------|----------------|----------------|-------------|
| | | runtime
s | noise
bit | runtime
s | noise
bit | $bits/s$ | | runtime
s | noise
bit | runtime
s | noise
bit | $bits/s$ |
| $p = 8088322049$ (33 bit) : | | | | | | | $p = 1096486890805657601$ (60 bit) : | | | | | |
| Iasta-3 | 32768 | 0.034 | 783 | 9.842 | 464 | 855 | 32768 | 0.027 | 756 | 9.804 | 274 | 1562 |
| Iasta-4 | 32768 | 0.030 | 783 | 9.797 | 386 | 430 | 32768 | 0.027 | 756 | 9.514 | 143 | 805 |
| Pasta-3 | 32768 | 0.033 | 783 | 22.639 | 339 | 186 | 32768 | 0.028 | 756 | 29.874 | 42 | 257 |
| Pasta-4 | 32768 | 0.054 | 783 | 10.8 | 214 | 97 | 65536 | 0.148 | 1611 | 61.767 | 680 | 31 |
| Masta-4 | 32768 | 0.057 | 783 | 28.336 | 223 | 149 | 65536 | 0.142 | 1611 | 147.318 | 695 | 52 |
| Masta-5 | 32768 | 0.055 | 783 | 20.625 | 99 | 102 | 65536 | 0.140 | 1611 | 110.527 | 491 | 35 |

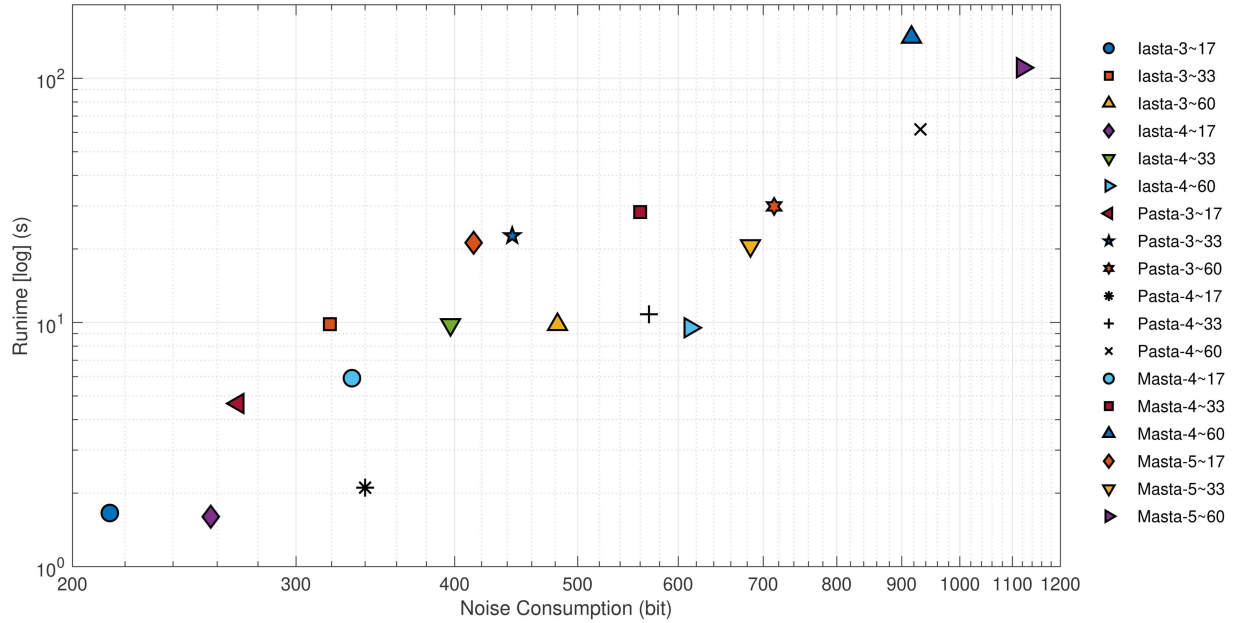


Fig. 6. Runtime and noise comparison of $\mathbb{F}_p$ ciphers for Homomorphic Conversion in SEAL (Security level $\lambda = 128$ bit).

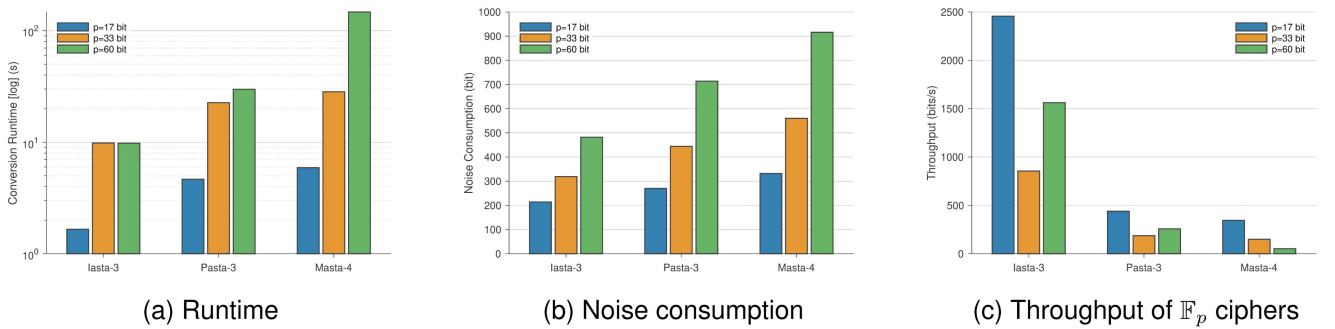


Fig. 7. Evaluation of Homomorphic Conversion Overhead and $\mathbb{F}_p$ Cipher Throughput at AND-depth 4.

TABLE VIII
COMPARISON OF RUNTIME AND THROUGHPUT BETWEEN IASTA, PASTA AND MASTA IN HELIB ($p = 17/33/60$ BIT)

| Cipher | n | 17-bit p | | | Throughput
$bits/s$ | n | 33-bit p | | | Throughput
$bits/s$ | n | 60-bit p | | | Throughput
$bits/s$ |
|---------|-------|------------|--------------|-----|------------------------|-------|------------|---------------|-----|------------------------|-------|------------|---------------|-----|------------------------|
| | | Enc | Conv | s | | | Enc | Conv | s | | | Enc | Conv | s | |
| Iasta-3 | 32768 | 0.031 | 11.748 | | 348 | 32768 | 0.037 | 12.343 | | 682 | 65536 | 0.094 | 40.530 | | 387 |
| Iasta-4 | 32768 | 0.034 | 7.561 | | 270 | 65536 | 0.074 | 23.693 | | 178 | 65536 | 0.108 | 34.867 | | 220 |
| Pasta-3 | 32768 | 0.029 | 13.432 | | 152 | 32768 | 0.032 | 14.409 | | 293 | 65536 | 0.089 | 45.668 | | 168 |
| Pasta-4 | 32768 | 0.031 | 5.638 | | 90 | 65536 | 0.075 | 16.908 | | 62 | 65536 | 0.088 | 23.500 | | 81 |
| Masta-4 | 32768 | 0.032 | 16.512 | | 124 | 65536 | 0.092 | 49.617 | | 85 | 65536 | 0.097 | 84.355 | | 91 |
| Masta-5 | 32768 | 0.038 | 14.030 | | 73 | 65536 | 0.087 | 38.516 | | 55 | 65536 | 0.132 | 61.827 | | 62 |

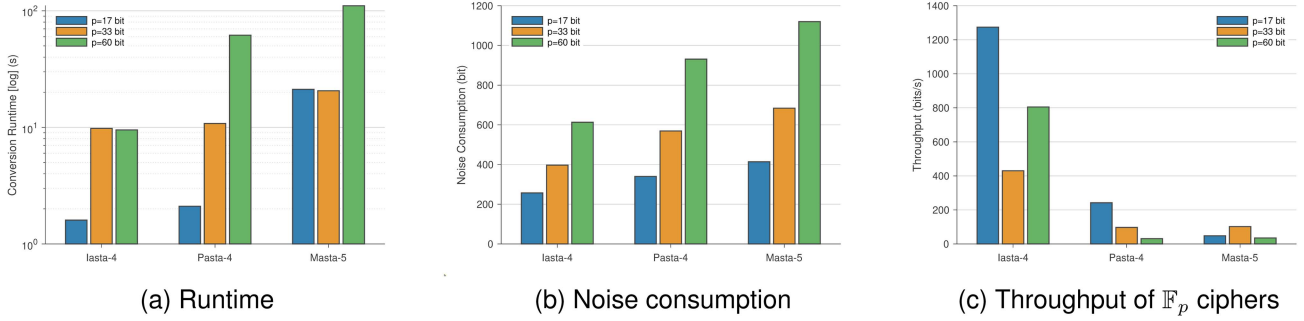


Fig. 8. Evaluation of Homomorphic Conversion Overhead and $\mathbb{F}_p$ Cipher Throughput at AND-depth 5.

advantage becomes more pronounced at larger p , where the gap between Iasta and its competitors widens further. Moreover, the design differences between the two libraries are also worth noting. SEAL provides efficient SIMD packing and conversion operations under the BFV scheme, along with highly efficient relinearization. In contrast, HELib incurs significant overhead for rotation operations, whereas SEAL handles rotations much more efficiently. Since Iasta involves many rotation operations, it benefits more from SEAL's efficient rotation implementation.

VI. CONCLUSION

This paper proposes Iasta, a novel HE-friendly cipher specifically designed for statistics and integer arithmetic in IoT scenarios. By employing polynomial modular arithmetic to align with the non-binary plaintext space of HE schemes, Iasta enables efficient packed evaluation operations when implemented on SEAL or Helib libraries. The affine layer of Iasta consists of rotation-based linear transformations, which significantly reduce client-side computational costs compared to state-of-the-art schemes. Meanwhile, the combination of Feistel S-boxes and a high-degree Cube S-box reduces required cipher rounds while effectively resisting recovery attacks on non-linear layers. Comprehensive cryptanalysis confirms that these modifications do not compromise the overall security of the cipher. Our unified implementation of Iasta, Pasta and Masta on the SEAL library demonstrates that Iasta achieves substantially lower client-side computation overhead, faster homomorphic conversion on servers, reduced noise consumption, and significantly higher throughput than prior solutions.

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